

Speech Recognition Spring 2025 Project 6

Due date: Feb 24th 2026 12pm Beijing time

In this assignment, you will continue by filling in the blanks to complete the decoding procedure of an end-to-end speech recognition system.

You are required to complete **Stage 5** in run.sh and report the **Word Error Rate (WER)** on the **test dataset** using the following configurations:

1. **CTC Decoding** (beam size: 1)
2. **CTC Decoding** (beam size: 20)
3. **Attention Decoding** (beam size: 1)
4. **Attention Decoding** (beam size: 20)
5. **Joint CTC-Attention Beam Decoding** (beam size: 20)

You can choose one model in project5 to report these 5 decoding results.

You may use the script tools/compute-wer.py to compute the WER. This is a fully functional utility borrowed from the WeNet toolkit for this purpose.

You should explain how you fill in the blanks in the report.

Bonus TASK (points 10)

Train and decode the model using **joint CTC and attention on aishell1 dataset**. Set ctc_weight in the config file as 0.5.

<https://www.openslr.org/33/>